

Counts or Conditional Percents?

AP Statistics · Unit 2 · Topic 2.1 · TEACHER KEY

CED TETHER

- **VAR-1.D** Identify questions to be answered about possible relationships in data. [Skill 1.A]
- **VAR-1.D.1** Apparent patterns and associations in data may be random or not.

Essential question: When group sizes differ, which comparison reveals whether two categorical variables are related?

DOK-3 skill: 4.B · **FRQ pattern:** counts-vs-conditional-percents-are-the-variables-related

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Supports Rosa using 70 percent versus 50 percent
- **reasoning-2** (required) — Explains that unequal group totals make 75 versus 35 misleading
- **reasoning-3** (required) — Limits the result to association, with no demonstrated cause

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Dividing 35 and 75 by the grand total instead of their row totals
- Saying 75 is a larger percentage because it is a larger count
- Claiming that association proves playing a sport increases sleep

[STAR] THE PROBLEM — annotated

Suppose a counselor records whether each of 200 students plays a school sport and whether the student usually gets at least 8 hours of sleep on a school night.

Kai: *Seventy-five nonathletes but only 35 athletes get at least 8 hours, so playing a sport goes with less sleep.*

Rosa: *Those counts compare groups of different sizes; 35 out of 50 may tell a different story from 75 out of 150.*

Usual sleep on a school night

Sport?	≥ 8 h	< 8 h	Total
Plays	35	15	50
Does not	75	75	150
Total	110	90	200

FIRST TAKE — one sentence before you work the parts

Which student's reasoning seems more useful for deciding whether sport participation and sleep are related? Name one table detail.

Key: Not graded. Expect the 75-to-35 count comparison to attract students before they normalize for group size.

[BOOK] COMPARE WITHIN GROUPS (keep on the page)

A **two-way table** records combinations of two categorical variables. A raw count answers *how many*; it does not by itself answer *how likely within a group*. To compare groups of different sizes, compute a **conditional relative frequency**: cell count $\div$ total for that row or column. If the conditional distributions differ, the variables appear associated. Association from observed data does not by itself establish cause.

DOK 1 (a) Name the two categorical variables.

Key: The variables are school-sport participation and usual school-night sleep category.

DOK 2 (b) Find the percent who get at least 8 hours of sleep in each sport group. Show the two fractions.

Key: Athletes: $35/50 = 70\%$. Nonathletes: $75/150 = 50\%$.

DOK 3 (c) Whose reasoning is better supported? Use both percentages and explain why comparing 75 with 35 can mislead. State whether this table shows that playing a sport causes more sleep. Write 3–4 sentences.

Key: Rosa's reasoning is supported: 70% of athletes versus 50% of nonathletes get at least 8 hours. The count 75 comes from a group of 150, three times the athlete group of 50, so the larger count does not mean a larger within-group percentage. The table shows an association for these students, but observing their choices does not show that playing a sport caused more sleep.